Monopsony in Motion

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Based on Burdett & Mortensen (1998) and Manning (2003, Ch. 2)

Model: Assumptions

- Continuum of m workers and 1 (unit measure of) firms; all **ex-ante identical**.
- Job offers arrive at Poisson rate λ for all workers (employed and unemployed).
- Jobs are destroyed at rate δ; define $k = \lambda/\delta$.
- Discount rate $= 0$ (firms maximize steady-state profits).
- Flow utility of unemployment: b (reservation wage).
- Revenue product of each employed worker: $p > b$.
- Workers accept any offer that beats their current wage.

Steady State: Unemployment and Wage Distribution

- **Unemployment stock** (flow in = flow out):

$$u \cdot \lambda = (m - u) \cdot \delta \implies u = \frac{m}{1 + k}$$

- Distribution of wages across firms: $F(w)$.
- **Wages must be dispersed:**
  - If a mass of firms offered the same wage $\hat{w}$, any one of them could gain workers by offering $\hat{w} + \varepsilon$.
  - Hence $F(w)$ is **continuous** (no mass points).

- Distribution of wages across workers: $G(w)$
- Steady state: Equal flow into and out of set of those with wages $\leq w$;

$$\underbrace{u \cdot \lambda \cdot F(w)}_{\text{Inflow}} = \underbrace{(m - u) \cdot G(w) \cdot [\delta + \lambda(1 - F(w))]}_{\text{Outflow}}$$

$$\implies G(w) = \frac{F(w)}{1 + k(1 - F(w))}$$

- Since $G(w) < F(w)$:
  Workers are **concentrated in higher-paying jobs** relative to the offer distribution.

Steady State: Firm Size

- The steady-state employment at a firm offering wage w satisfies

$$(m-u)\frac{G'(w)}{F'(w)} = (m-u)\frac{\partial G(w)}{\partial F(w)}.$$

- Thus:

$$N(w) = \frac{m \cdot k}{\left[1 + k(1 - F(w))\right]^2}.$$

- **Intuition:**
  - Higher-wage firms lose workers more slowly (lower separation rate).
  - Higher-wage firms recruit from a larger pool (more workers currently earning less).
  - Both effects increase firm size $\implies$ **firm size is increasing in wages**.

Equilibrium Wage Distribution

- Firm profits: $\pi(w) = (p - w) \cdot N(w)$
- **Profit equalization:**
  In equilibrium, every offered wage must yield the same profit π^*.
- Setting $\pi(w) = \pi(b)$ for all w in the support of F:

$$(p - w) \cdot \frac{1}{[1 + k(1 - F(w))]^2} = (p - b) \cdot \frac{1}{(1 + k)^2}$$

- Solving for $F(w)$ (distribution of wages across firms):

$$F(w) = \frac{1 + k}{k} \left(1 - \sqrt{\frac{p - w}{p - b}} \right), \qquad w \in [b, \bar{w}].$$

- Solving for $G(w)$ (distribution of wages across workers):

$$\boxed{G(w) = \frac{1}{k}\left(\sqrt{\frac{p-b}{p-w}} - 1\right)}$$

- The upper bound on wages is $\bar{w} = p - \left(\frac{1}{1+k}\right)^2 (p-b) < p$.
- The **expected wage** is $E(w) = \frac{1}{1+k}\, b + \frac{k}{1+k}\, p$ — a weighted average of b and p.

Comparative statics

- As $k \to \infty$ ($\lambda \gg \delta$): $G(w) \to \mathbf{1}(w \geq p)$, $u \to 0$: **competitive equilibrium**
- Higher outside option $b \implies$ higher wages, **no change in unemployment**
- A minimum wage $w^{\min}$ has the same effect as raising b
- Setting either b or $w^{\min}$ equal to p **maximizes worker welfare**

Key Conceptual Insights

1. **Wage dispersion without heterogeneity:** All firms are identical yet offer different wages. Dispersion is an equilibrium phenomenon driven by search frictions.
2. **Wages below marginal product:** Despite a continuum of identical competitive firms, all wages lie strictly below p. Monopsony power arises from frictions, not market concentration.
3. **Firm size–wage correlation:** Larger firms pay higher wages — consistent with the well-documented employer size–wage effect.
4. **Multiple determinants of wages:** Beyond productivity p, wages depend on b (outside option), λ (offer arrival rate), and δ (job destruction rate).
5. **Wage inequality from luck:** Workers with identical productivity end up in different jobs purely by chance.

Worker Heterogeneity

- Allow reservation wages to vary: $b \sim H(b)$.
- Conditional on b, this is the same model as before.
- Labor supply to firms offering wage w from workers with reservation wage $b \leq w$ is given by

$$N(w|b) = \frac{mk}{(1 + k \cdot (1 - F(w)))^2} dH(b),$$

and thus

$$N(w) = \int_{\underline{b}}^{w} N(w|b)db = \frac{mkH(w)}{(1 + k \cdot (1 - F(w)))^2}.$$

- Profit equalization now requires:

$$\frac{(p-w)\,H(w)}{(p-\underline{w})\,H(\underline{w})} = \frac{[1+k(1-F(w))]^2}{(1+k)^2}$$

- Solving:

$$F(w) = \frac{1+k}{k}\left[1-\left(\frac{(p-w)H(w)}{(p-\underline{w})H(\underline{w})}\right)^{1/2}\right]$$

- The lowest offered wage $\underline{w}$ maximizes $(p-w)H(w)$ — the firm trades off profit margin against the pool of available workers.

Employer Heterogeneity

- Now suppose employers differ in productivity. Two types: $p_2 > p_1$.
- **Key result:** More productive firms offer higher wages.

$$F_i(w) = \frac{1+k}{k}\left[1 - \sqrt{\frac{p_i - w}{p_i - \underline{w}_i}}\right], \quad i = 1, 2$$

- with $\bar{w}_1 = \underline{w}_2$ (the wage distributions are *stacked*).

Implications:

- More productive firms have larger workforces **and** higher wages.
- Productivity differences + search frictions explain persistent **cross-firm and inter-industry wage differentials**.

Policy Implications

Minimum wage:

- Raises the lower bound of the wage distribution
- Shifts the entire equilibrium distribution upward (stochastic dominance)
- Can **increase employment** by reducing monopsonistic distortions
- Setting $w^{\min} = p$ maximizes worker welfare in the basic model

Outside options:

- Improving b (e.g., unemployment benefits, basic income, job guarantee) raises wages **without affecting unemployment**
- Policies that increase λ (better job matching, information) increase competition among employers and push wages toward p

Summary

Feature	Competitive model	Burdett–Mortensen
Wage dispersion	Only from heterogeneity	Equilibrium outcome
w vs. p	$w = p$	$w < p$ always
Firm size–wage	No relation	Positive
Min. wage effect	Employment $\downarrow$	Employment $\uparrow$ or $\approx$

The Burdett–Mortensen model provides:

- a unified, internally consistent framework
- explaining wage dispersion, employer size–wage effects, and the effects of minimum wages and outside options
- from two simple assumptions: frictions $+$ wage-posting by firms.

Thank you!